
Lecture 7-8: Interferometers and Aperture Synthesis 2

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Radio Interferometer

The visibility function (note the different definition in "tools") is :

$$\begin{aligned}\mathcal{V}(\mathbf{D}_\lambda) &= \int_{4\pi} A(\boldsymbol{\sigma})B(\boldsymbol{\sigma})e^{-i2\pi\mathbf{D}_\lambda \cdot \boldsymbol{\sigma}} d\Omega \\ &= \int_{4\pi} A(\boldsymbol{\sigma})B(\boldsymbol{\sigma}) \cos(2\pi\mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega \\ &\quad -i \int_{4\pi} A(\boldsymbol{\sigma})B(\boldsymbol{\sigma}) \sin(2\pi\mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega \\ &\equiv V_R + iV_I\end{aligned}$$

The 1st term is the [cosine or even or real part](#), while the 2nd term the [sine or odd or imaginary part](#) of the visibility. Both terms can be measured simultaneously with a two-correlator device called the [complex correlator](#), with

$$\begin{aligned}V_R &= \int_{4\pi} A(\boldsymbol{\sigma})B(\boldsymbol{\sigma}) \cos(2\pi\mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega \\ V_I &= - \int_{4\pi} A(\boldsymbol{\sigma})B(\boldsymbol{\sigma}) \sin(2\pi\mathbf{D}_\lambda \cdot \boldsymbol{\sigma}) d\Omega\end{aligned}$$

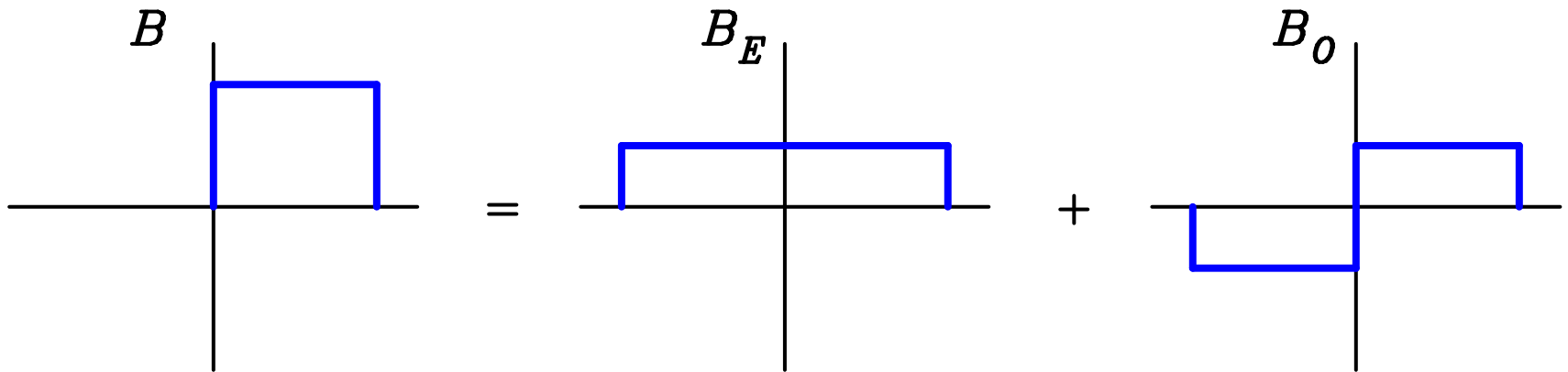
$$\tan \phi_v = \frac{V_I}{V_R} \quad \text{and} \quad |\mathcal{V}|^2 = V_R^2 + V_I^2$$

Radio Interferometer

Any real function, B , can be expressed as the sum of two real functions which have specific symmetries:

An even part: $B_E(\sigma) = [B(\sigma) + B(-\sigma)]/2 = B_E(-\sigma)$

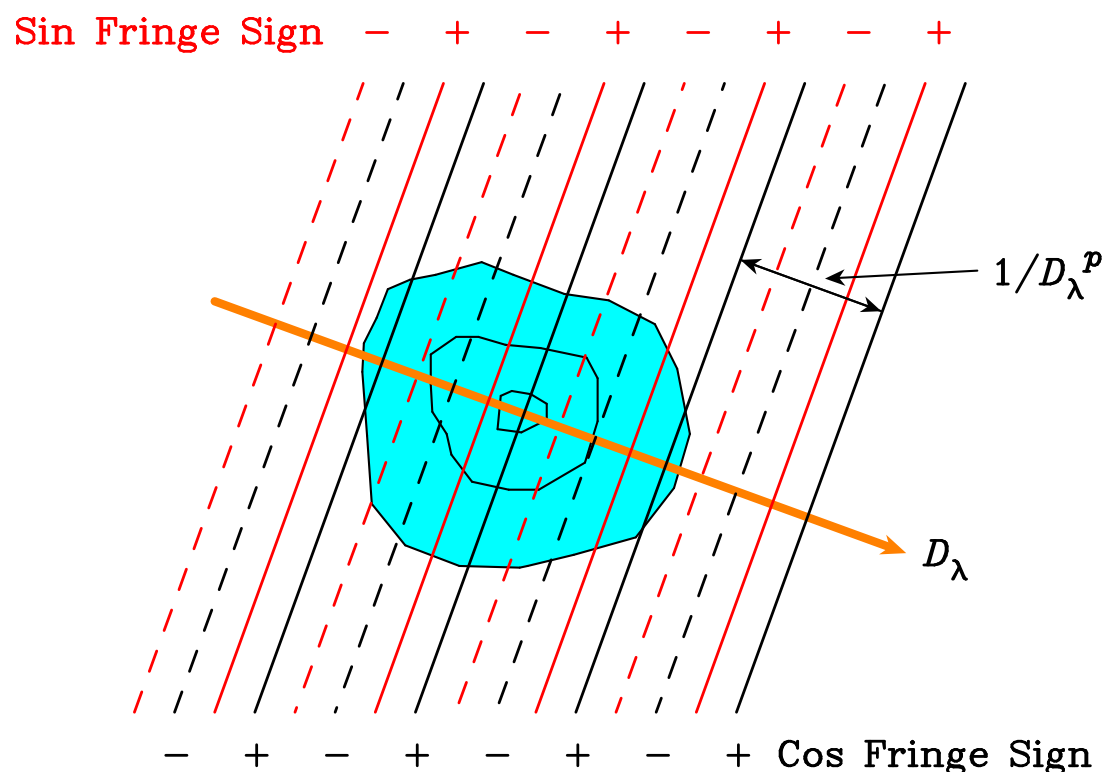
An odd part: $B_O(\sigma) = [B(\sigma) - B(-\sigma)]/2 = -B_O(-\sigma)$



Thus, the visibility needs to have two parts, the even and the odd parts.

Radio Interferometer

The correlator can be thought of "casting" two sinusoidal fringe patterns of angular scale $1/D_\lambda^p$ radians, onto the sky, within the primary telescope beam. D_λ^p : projected baseline. The correlator multiplies the source brightness by these wave patterns, and integrates the result over the primary beam.



Fringe separation: $1/D_\lambda^p$ radian. Averaged signal is dependent on D_λ^p (or baseline projected to the sky), a function of direction.

Choosing appropriate Coor. System

To solve the visibility function, choose an appropriate Coordinate System (u, v, w) v.s. (ξ, η, ζ)

$$\mathcal{V}(\mathbf{D}_\lambda) = \int_{4\pi} A(\boldsymbol{\sigma}) B(\boldsymbol{\sigma}) e^{-i2\pi \mathbf{D}_\lambda \cdot \boldsymbol{\sigma}} d\Omega$$

$$\mathbf{D}_\lambda = (u, v, w)$$

$$\mathbf{s} = (\xi, \eta, \zeta)$$

$$\mathbf{s}_0 = (0, 0, 1)$$

$$\boldsymbol{\sigma} = \mathbf{s} - \mathbf{s}_0 = (\xi, \eta, \zeta - 1)$$

$$d\Omega = \frac{d\xi d\eta}{\zeta}$$

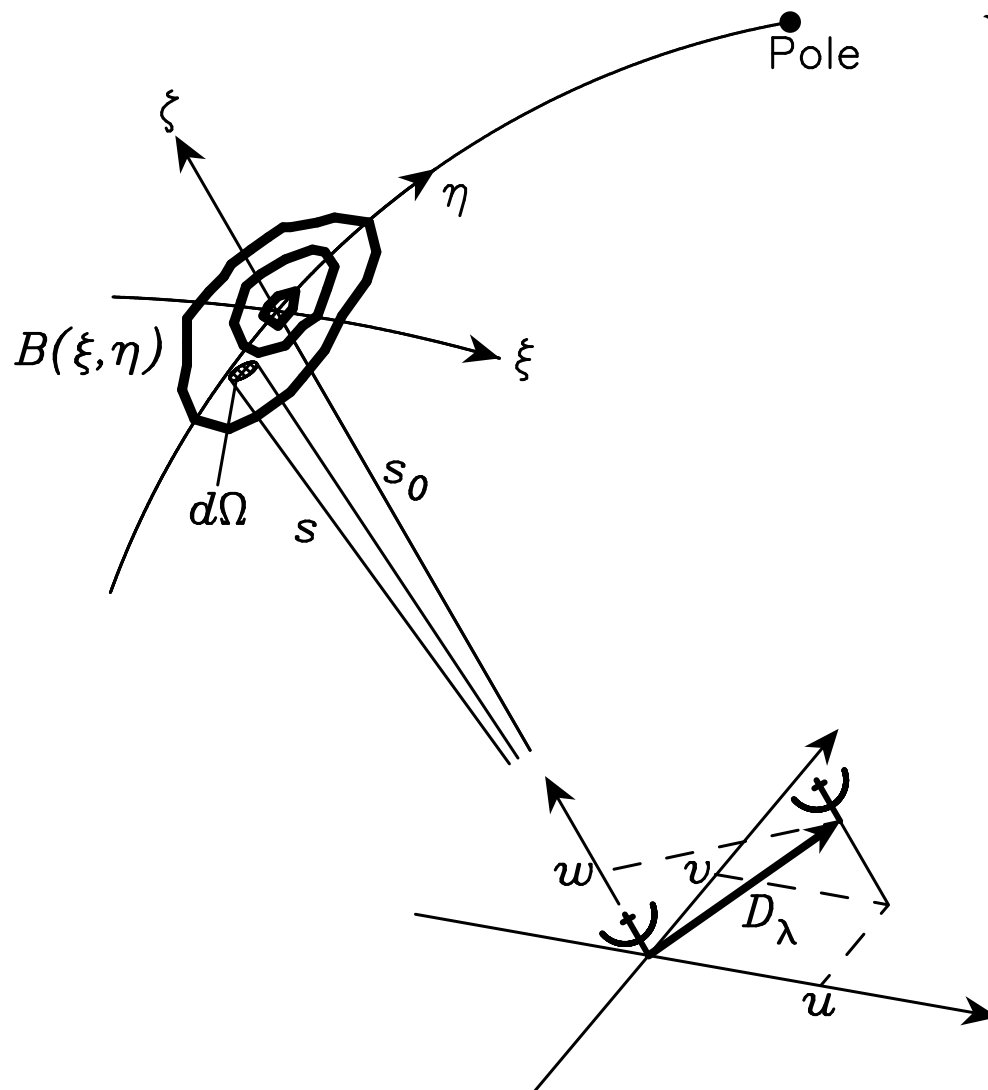
$w \rightarrow \mathbf{s}_0$, then $u \rightarrow$ east, $v \rightarrow$ north

Thus, center of $(u, v, w) \rightarrow \mathbf{s}_0$

and (u, v, w) rotates as the earth rotates.

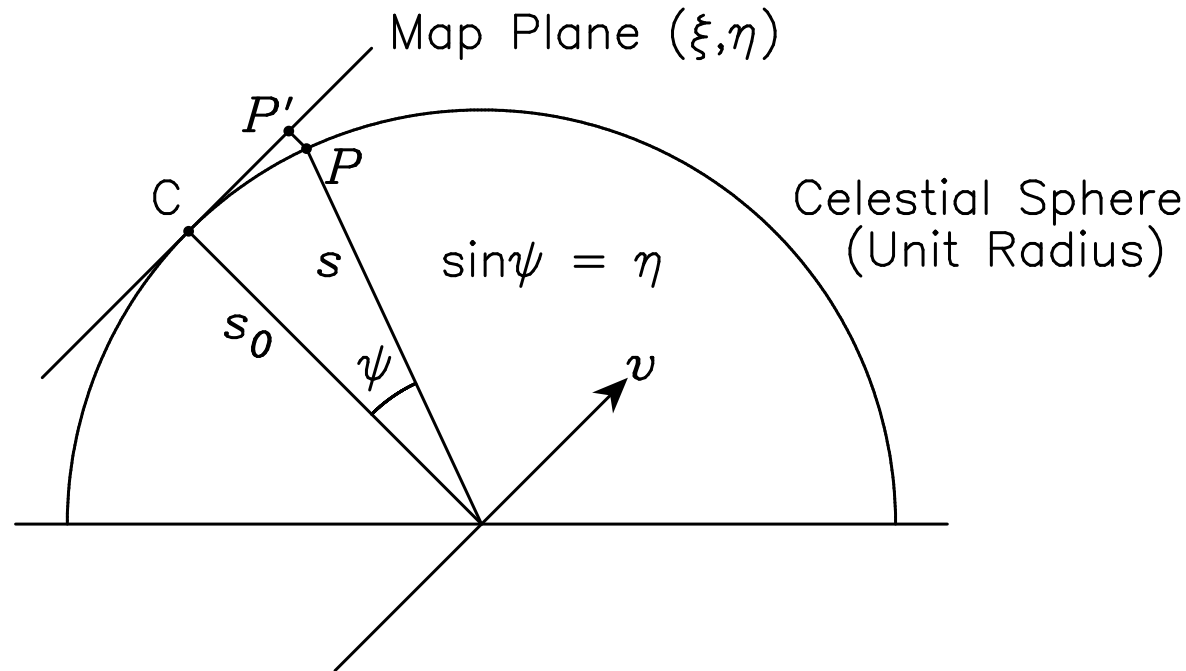
ξ, η, ζ : direction cosines of $\mathbf{s}$ on u, v and w .

$$\xi^2 + \eta^2 + \zeta^2 = 1, \zeta = \sqrt{1 - \xi^2 - \eta^2}$$



Radio Interferometer

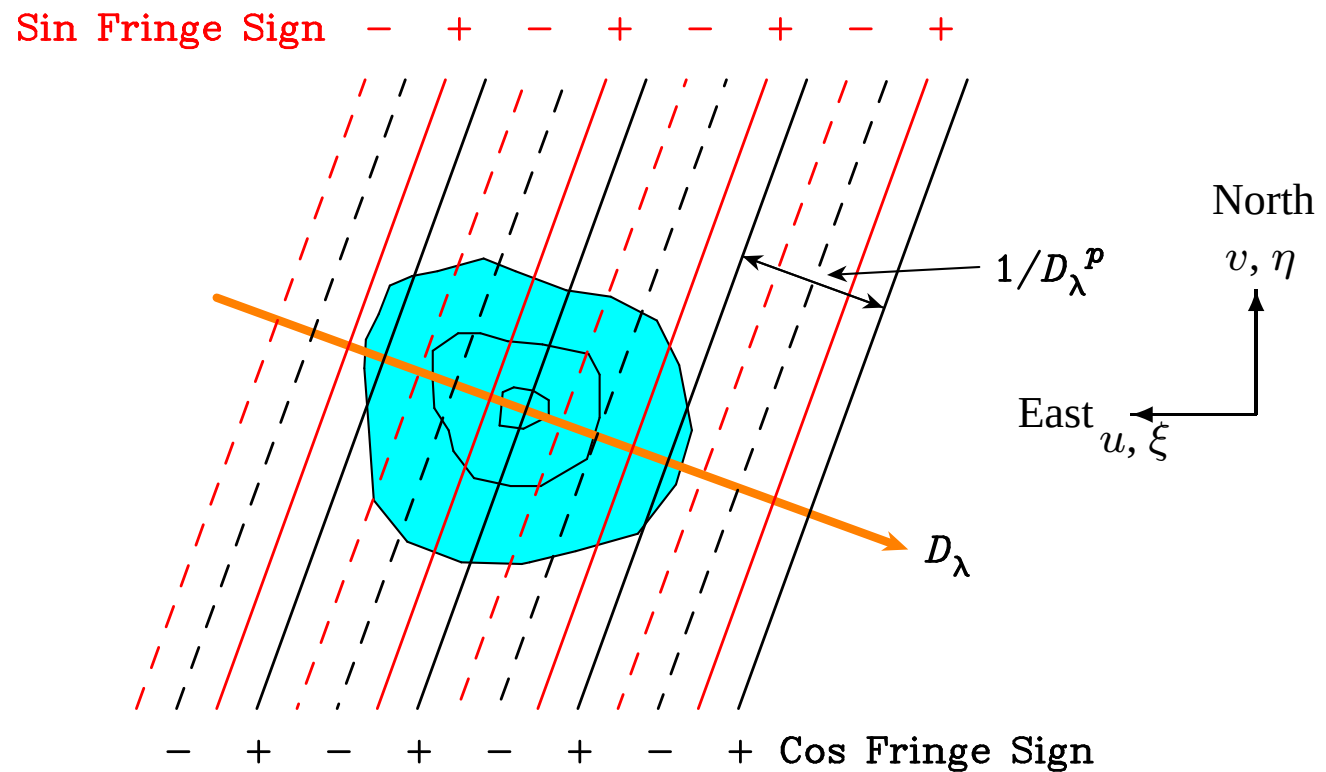
How do we define the direction cosines? E.g., in v axis:



Mapping of the celestial sphere onto a plane, shown in one dimension. The position P is measured in terms of the direction cosine η w.r.t the v axis. When projected onto a plane surface with a scale linear in η , P appears at P' at a distance from the field center C proportional to η .

Radio Interferometer

With the chosen coordinate system, the plane of the sky is then



Radio Interferometer

With the appropriate coordinate system, we have

$$\mathcal{V}(u, v, w) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} A(\xi, \eta) B(\xi, \eta) \exp\{-i2\pi[u\xi + v\eta + w(\sqrt{1 - \xi^2 - \eta^2} - 1)]\} \\ \times \frac{d\xi d\eta}{\sqrt{1 - \xi^2 - \eta^2}}$$

with $A(\xi, \eta) = 0$ for $\xi^2 + \eta^2 \leq 1$. This is OK since A drops rapidly when $\xi^2 + \eta^2 > l^2$, where l is the full width of the primary beam of the antenna and $l^2 \ll 1$. If ξ and η are small enough, we have

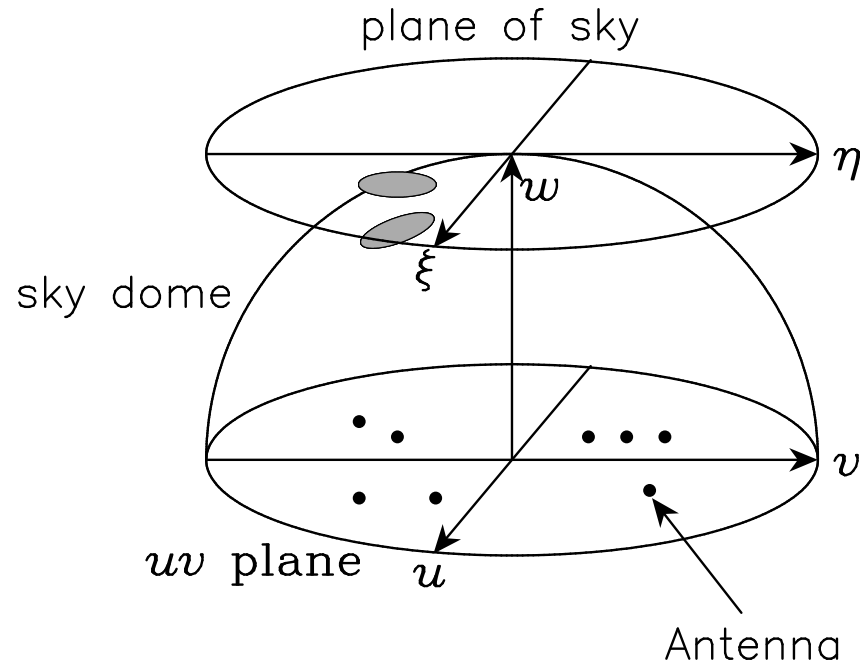
$$w(\sqrt{1 - \xi^2 - \eta^2} - 1) \simeq -\frac{1}{2}(\xi^2 + \eta^2)w$$

i.e, higher order of ξ and η , and thus can be neglected. Then

$$\mathcal{V}(u, v, w) \simeq \mathcal{V}(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{A(\xi, \eta) B(\xi, \eta)}{\sqrt{1 - \xi^2 - \eta^2}} e^{-i2\pi(u\xi + v\eta)} d\xi d\eta$$

Thus for a restricted range of ξ and η , $\mathcal{V}(u, v, w)$ is approximately independent of w , and can be considered to be projected on the flat uv plane.

Radio Interferometer



The sky dome and its approximation as a flat "plane of the sky" are shown. For baselines with a significant w component, the approximation of a flat sky causes **distortions near the edges of the map**. The gray oval on the plane of the sky is really on the curved sky dome. The approximation introduces a phase error equal to 2π times the neglected term, or $\pi(\xi^2 + \eta^2)w$. The maximum $w \sim (\mathbf{D}_\lambda)_{\max}$ and the synthesized beam $\theta_b \sim (\mathbf{D}_\lambda)_{\max}^{-1}$. So the **max. phase error** is $\phi_{\text{err}} \sim \pi(\theta_f/2)^2 \theta_b^{-1}$, where θ_f is the field of view. For the SMA at 345 GHz, $\theta_f \sim 36''$ and $\theta_b \sim 1''$, then $\phi_{\text{err}} \sim 0.0016$ radian (0.3 deg). Set this error to less than 0.1 radian (~ 6 deg), we have $\theta_f < \frac{1}{3}\sqrt{\theta_b} \sim 2.5$ arcmin. As seen later, a phase shift ϕ_{err} in $\mathcal{V}$ is related to position shift θ_{err} in the map, i.e., $2\pi u \theta_{\text{err}} = \phi_{\text{err}}$. Let $u \sim (\mathbf{D}_\lambda)_{\max} \sim \theta_b^{-1}$, then $\theta_{\text{err}} \sim \frac{\phi_{\text{err}}}{2\pi} \theta_b \sim \frac{1}{8} \theta_f^2$.

Radio Interferometer

Thus for a restricted range of ξ and η , $\mathcal{V}(u, v, w)$ is approximately independent of w , and for the inverse transform we can write

$$\frac{A(\xi, \eta)B(\xi, \eta)}{\sqrt{1 - \xi^2 - \eta^2}} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{V}(u, v) e^{i2\pi(u\xi + v\eta)} du dv$$

and then

$$B(\xi, \eta) = \frac{\sqrt{1 - \xi^2 - \eta^2}}{A(\xi, \eta)} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{V}(u, v) e^{i2\pi(u\xi + v\eta)} du dv$$

However, the uv plane is not fully sampled. The coverage in the uv plane is sampled by available baselines. Thus, introducing a **sampling function** $S(u, v)$, we have a "dirty image":

$$B^D(\xi, \eta) = \frac{\sqrt{1 - \xi^2 - \eta^2}}{A(\xi, \eta)} \sum_k \mathcal{V}(u_k, v_k) S(u_k, v_k) e^{i2\pi(u_k \xi + v_k \eta)}$$

with the sampling function

$$S(u_k, v_k) = \delta(u - u_k, v - v_k)$$

Radio Interferometer

So what we have is the "dirty image" obtained via a **discrete Fourier transform (DFT)**:

$$B^D(\xi, \eta) = \frac{\sqrt{1 - \xi^2 - \eta^2}}{A(\xi, \eta)} \sum_k \mathcal{V}(u_k, v_k) S(u_k, v_k) e^{i2\pi(u_k \xi + v_k \eta)}$$

Since $F(s)G(s) \rightarrow f(x) \otimes g(x)$, we have

$$B^D(\xi, \eta) = B(\xi, \eta) \otimes b(\xi, \eta)$$

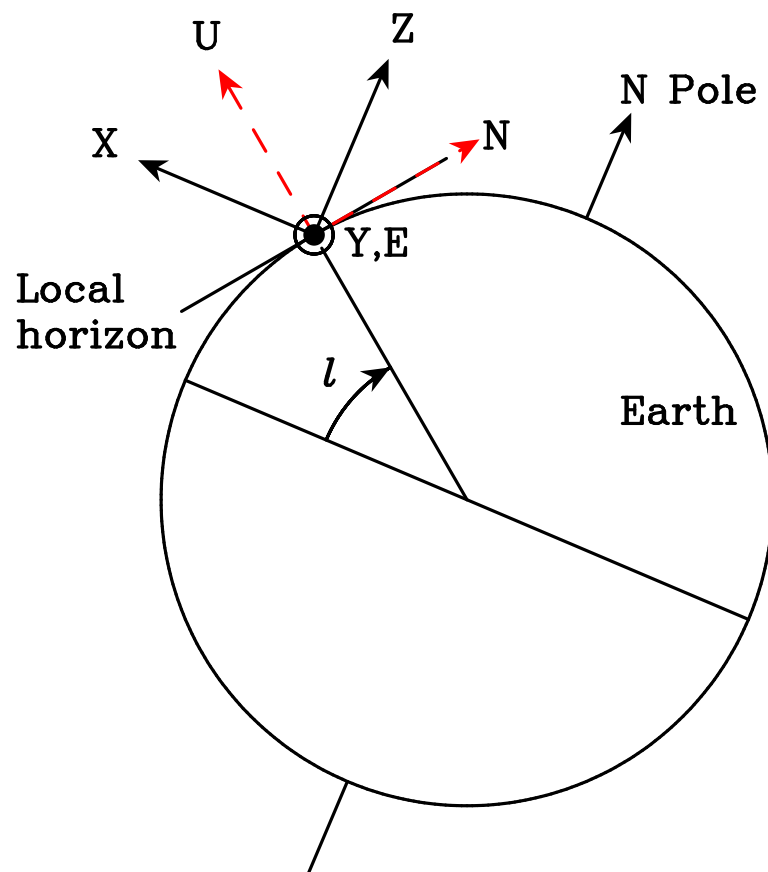
where b is the dirty beam (PSF, point spread function) given by

$$b(\xi, \eta) = \sum_k S(u_k, v_k) e^{i2\pi(u_k \xi + v_k \eta)}$$

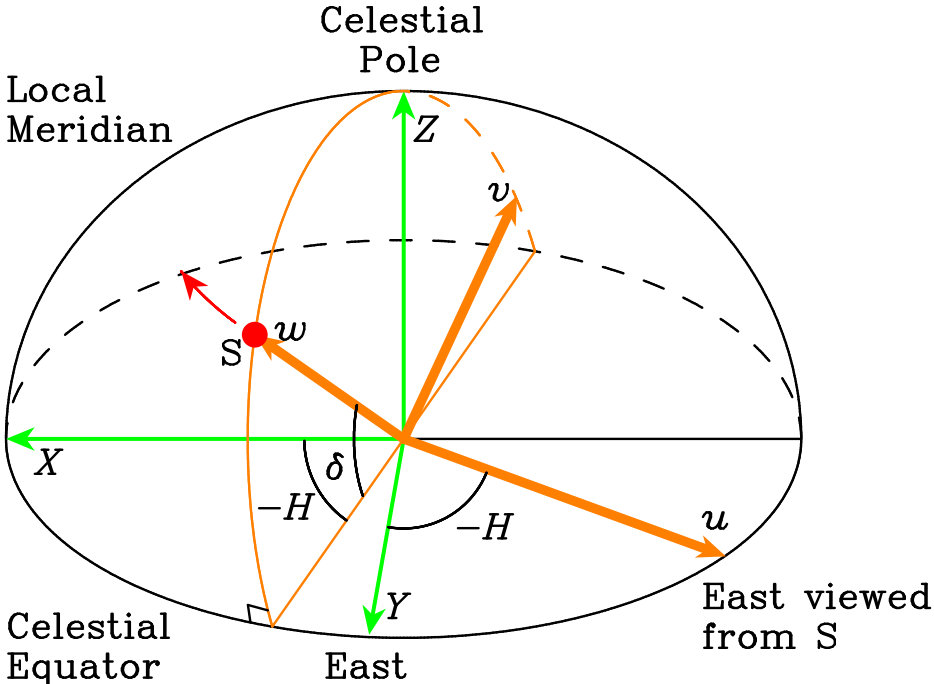
Thus, what we really have is **the image brightness convolved with the dirty beam**. To retrieve the image brightness, we need **deconvolution**. The main lobe of this dirty beam can be fitted by a Gaussian beam and is called the synthesized beam. **This synthesized beam determines the angular resolution and its size depends on $\lambda/D_{\max}$** . Assuming $\frac{\sqrt{1 - \xi^2 - \eta^2}}{A(\xi, \eta)} = 1$, the dirty beam is also the dirty image we would obtain if we observed a unit point source at the phase center, for which $\mathcal{V} = 1 + 0i$,

Antenna Spacing Coordinate

Antennas are typically measured along the ground. We will use a right-handed coordinate system of **East, North, UP** (E, N, U). These coordinates are relative to the local horizon, however, and will change depending on where we are on the spherical Earth. It is convenient to use a coordinate system aligned with the Earth's rotational axis, for which we will use coordinates (X, Y, Z). Conversion between these systems is done via a simple rotation matrix:

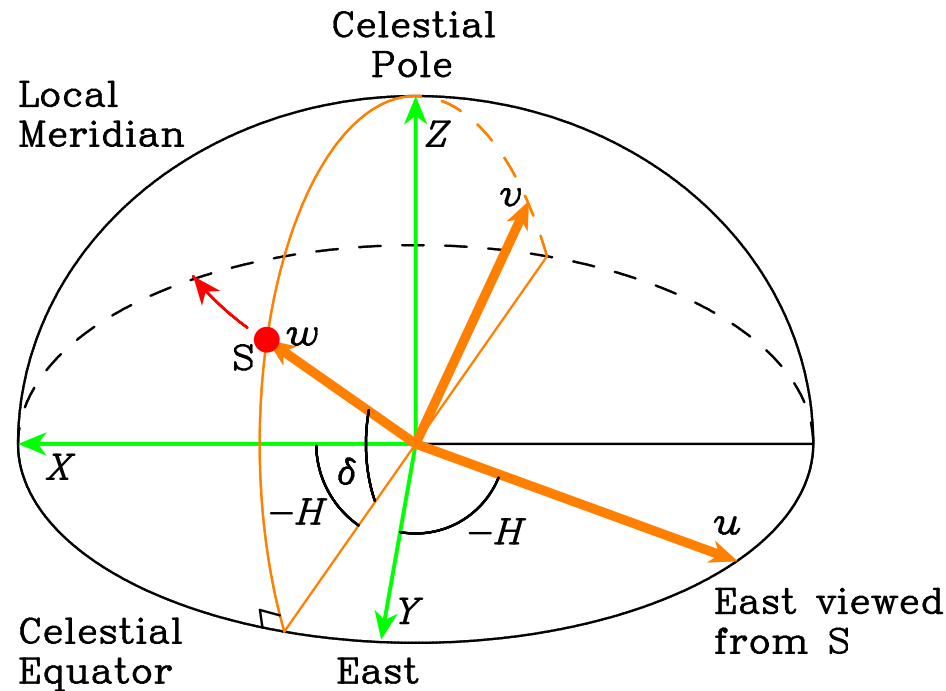


$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0 & -\sin l & \cos l \\ 1 & 0 & 0 \\ 0 & \cos l & \sin l \end{bmatrix} \begin{bmatrix} E \\ N \\ U \end{bmatrix}$$



Here (X, Y, Z) is a right-handed Cartesian coordinate system used to specify the relative positions of the antennas in the array. X and Y are measured in a plane parallel to the earth's equator, X in the meridian plane (defined as the plane through the poles of the earth, the zenith, and the reference point in the array), Y toward the west, and Z is measured toward the north pole. In terms of hour angle H and declination δ , (X, Y, Z) are measured toward $(H = 0, \delta = 0)$, $(H = -6^{\text{h}}, \delta = 0)$, and $(\delta = 90^\circ)$, respectively. As shown, when the source S rises in the eastern hemisphere, H is negative. Here, celestial pole is just the earth's pole extended into space, celestial equator is the earth's equator. Source rises in the east, passes through the local meridian and then sets in the west.

Antenna Spacing Coordinate



Once (X, Y, Z) are measured for given baseline, the components (u, v, w) are given by

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = R_x(\delta)R_z(-H) \begin{bmatrix} X_\lambda \\ Y_\lambda \\ Z_\lambda \end{bmatrix} = \begin{bmatrix} \sin H & \cos H & 0 \\ -\sin \delta \cos H & \sin \delta \sin H & \cos \delta \\ \cos \delta \cos H & -\cos \delta \sin H & \sin \delta \end{bmatrix} \begin{bmatrix} X_\lambda \\ Y_\lambda \\ Z_\lambda \end{bmatrix}$$

Note that u is in the XY plane. H changes as the earth rotates, producing different (u, v, w) .

Example: BIMA

Size: 6 meter; Number of antennas: $n = 10$; Number of baselines: $n(n - 1)/2 = 45$.

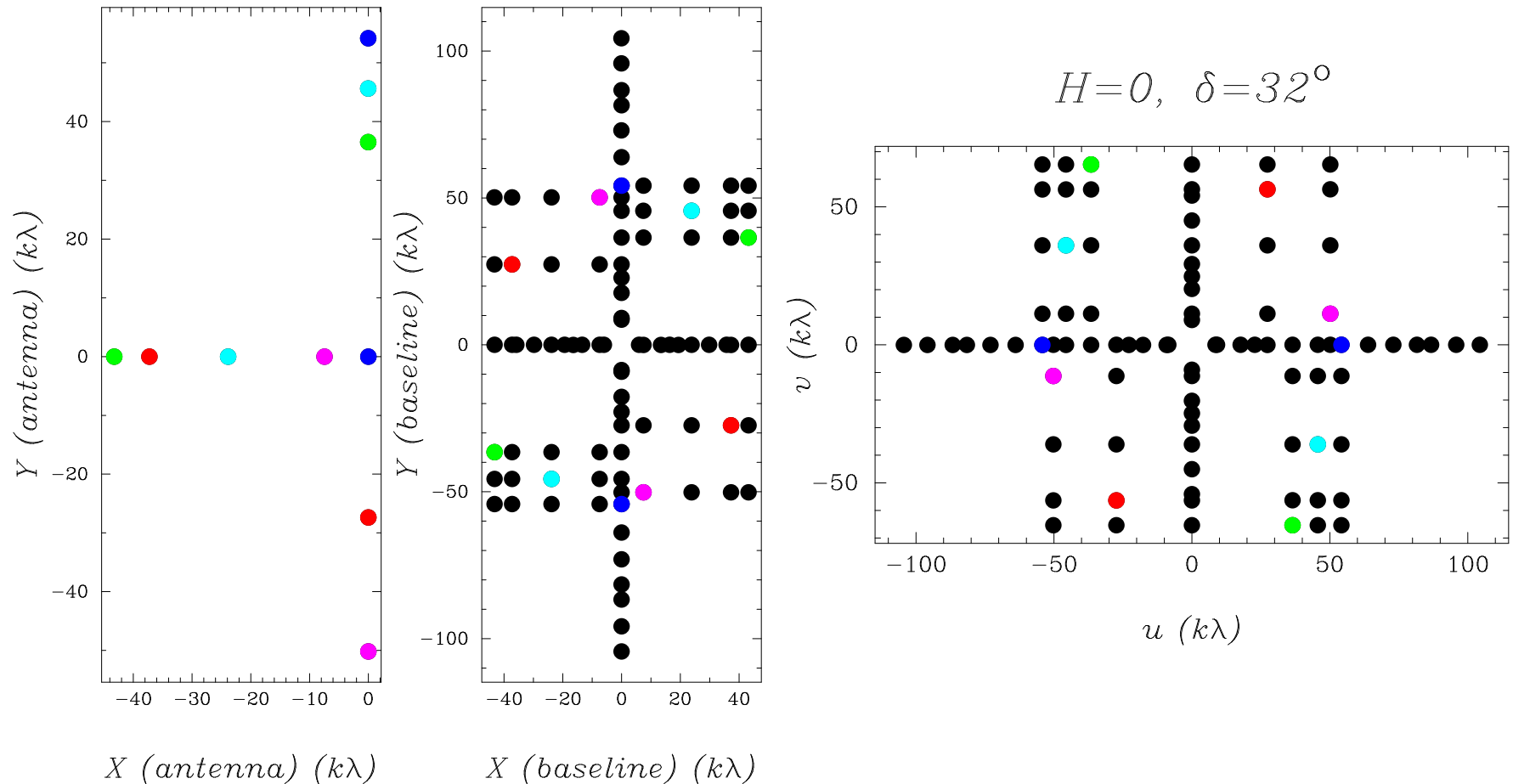


BIMA no longer exists, it merged with OVRO and is part of CARMA.



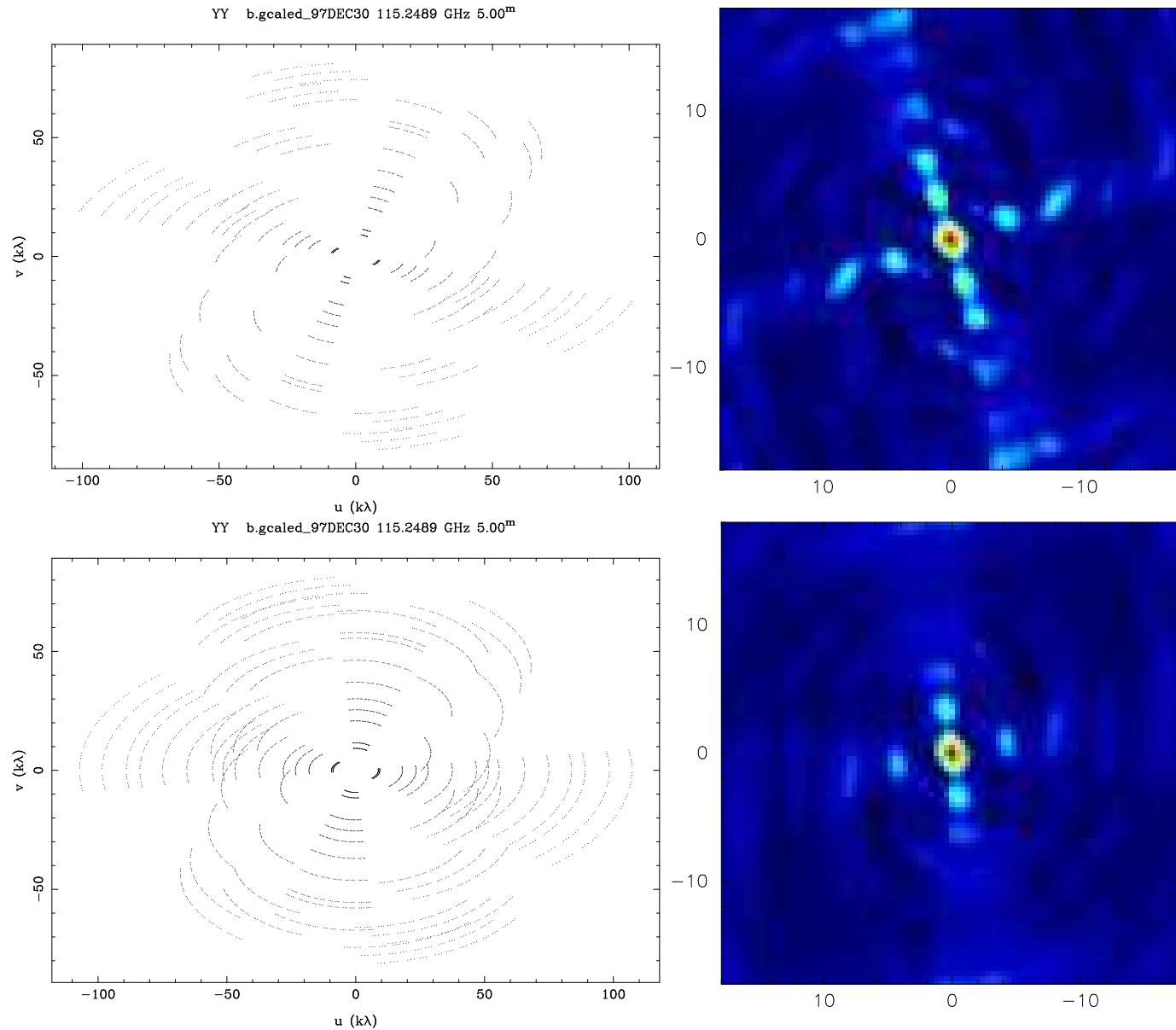
Example: BIMA Zenith snapshot

Number of antennas: $n = 10$, number of baselines: $n(n - 1)/2 = 45$.



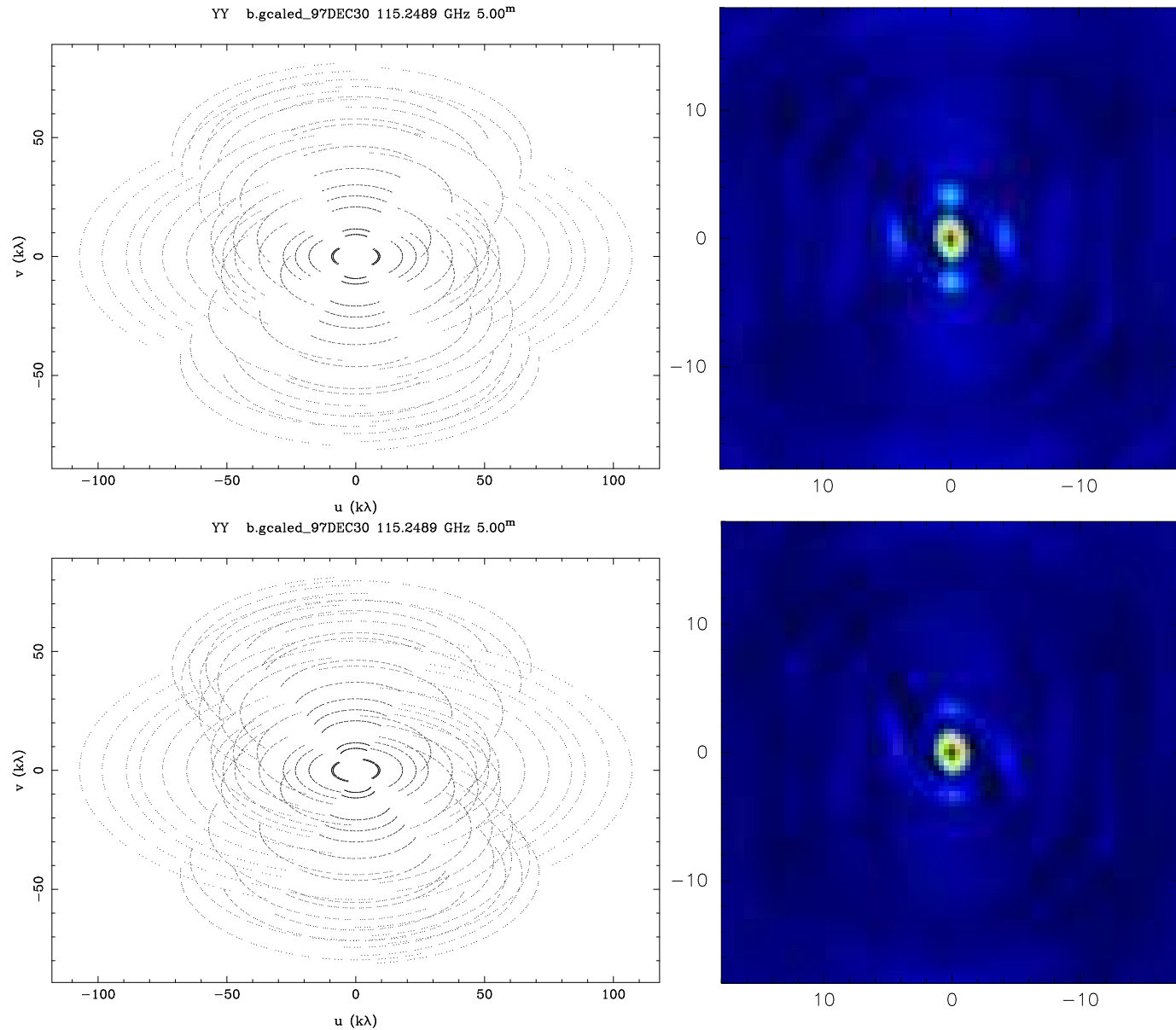
Since $\mathcal{V}(-u, -v) = \mathcal{V}^*(u, v)$, any observation supplies simultaneously measurements on two uv points. At $H = 0$, $u \rightarrow -X_\lambda \sin \delta$ and $v \rightarrow Y_\lambda$.

Example: BIMA

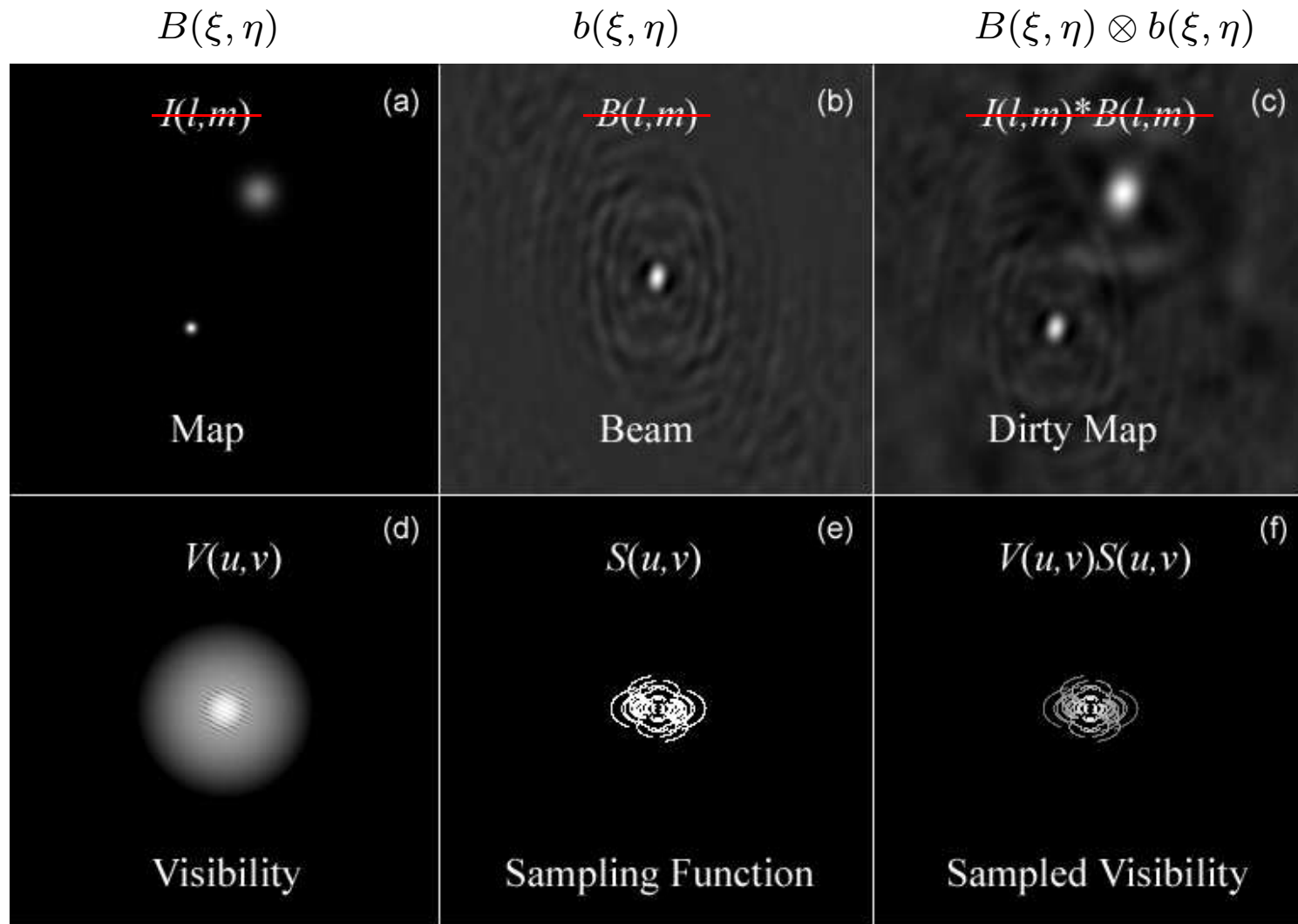


Traditionally, u increases to the right, while ξ increases to the left!

A point source or dirty beam



Example: Two Gaussians

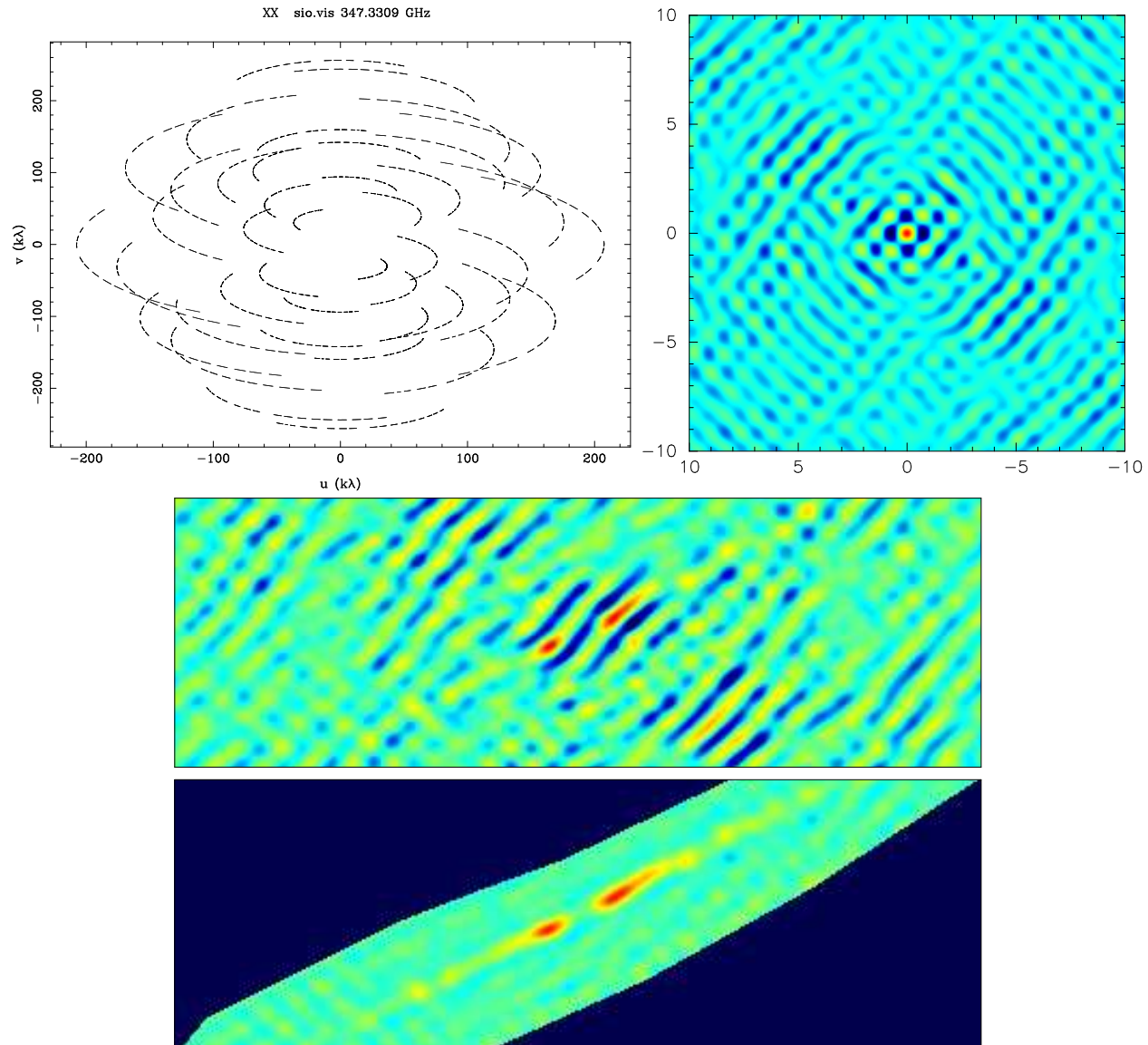


Example: SMA



Submillimeter Array: Number of antennas: $n = 8$, number of baselines: $n(n - 1)/2 = 28$.

Example: SMA HH 211 Jet

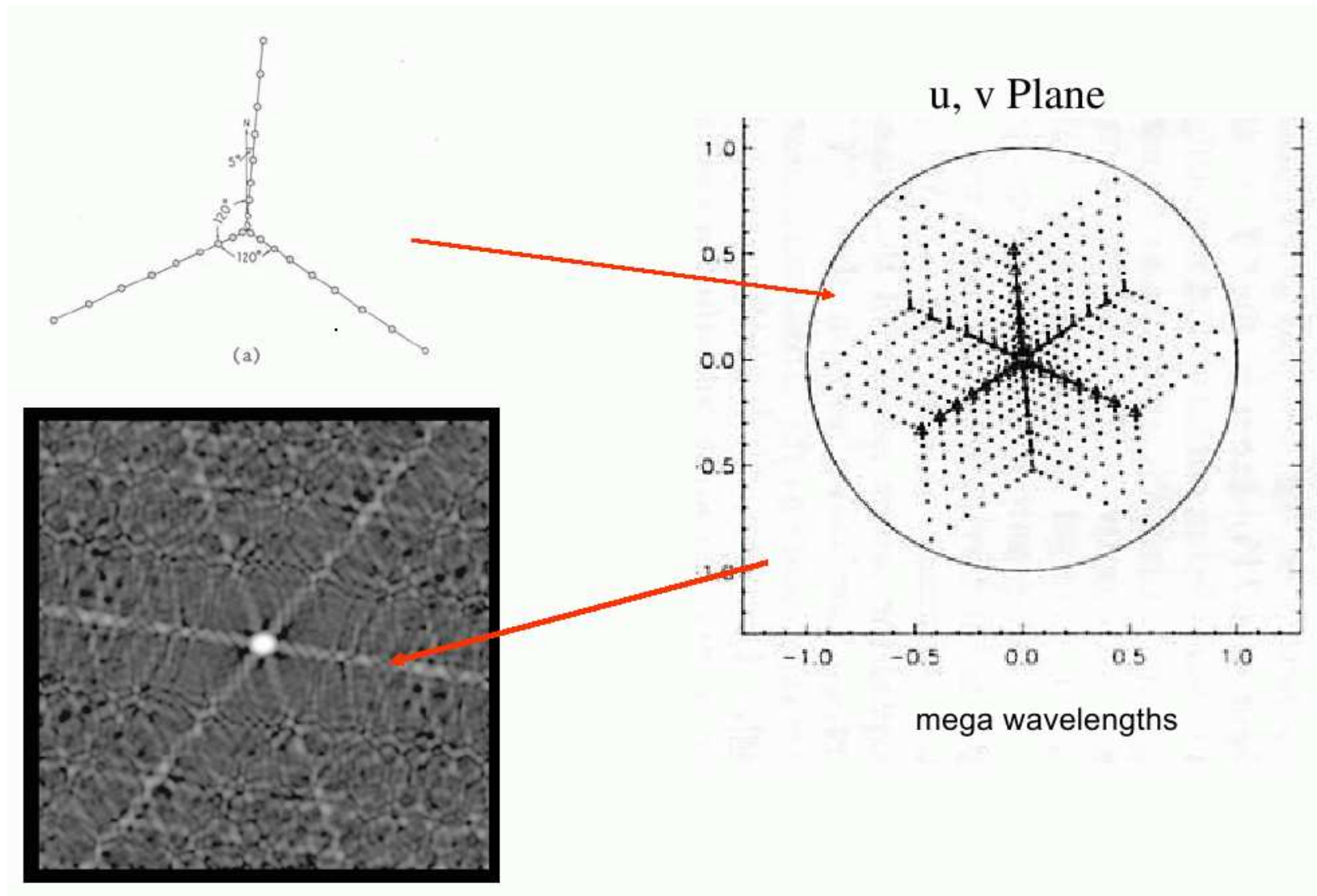


Example: VLA

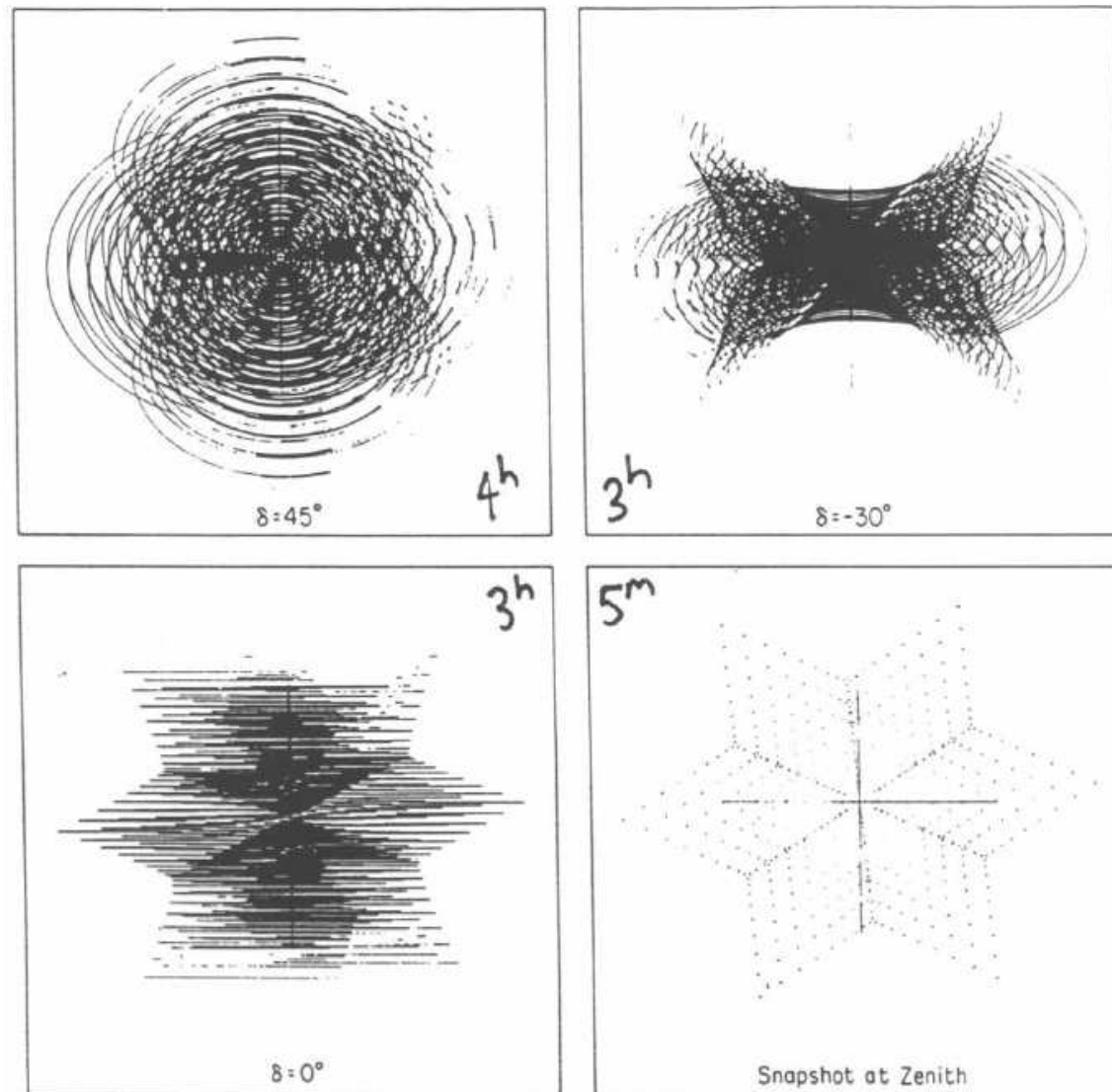


The Very Large Array (VLA) consists of 27 antennas in a Y-shaped configuration on the Plains of San Agustin, fifty miles west of Socorro, New Mexico. Each antenna is 25 meters in diameter. The data from the antennas is combined electronically to give the resolution of an antenna 36 km across, with the sensitivity of a dish 130 (i.e., $\sqrt{27 \times 25^2}$) meters in diameter.

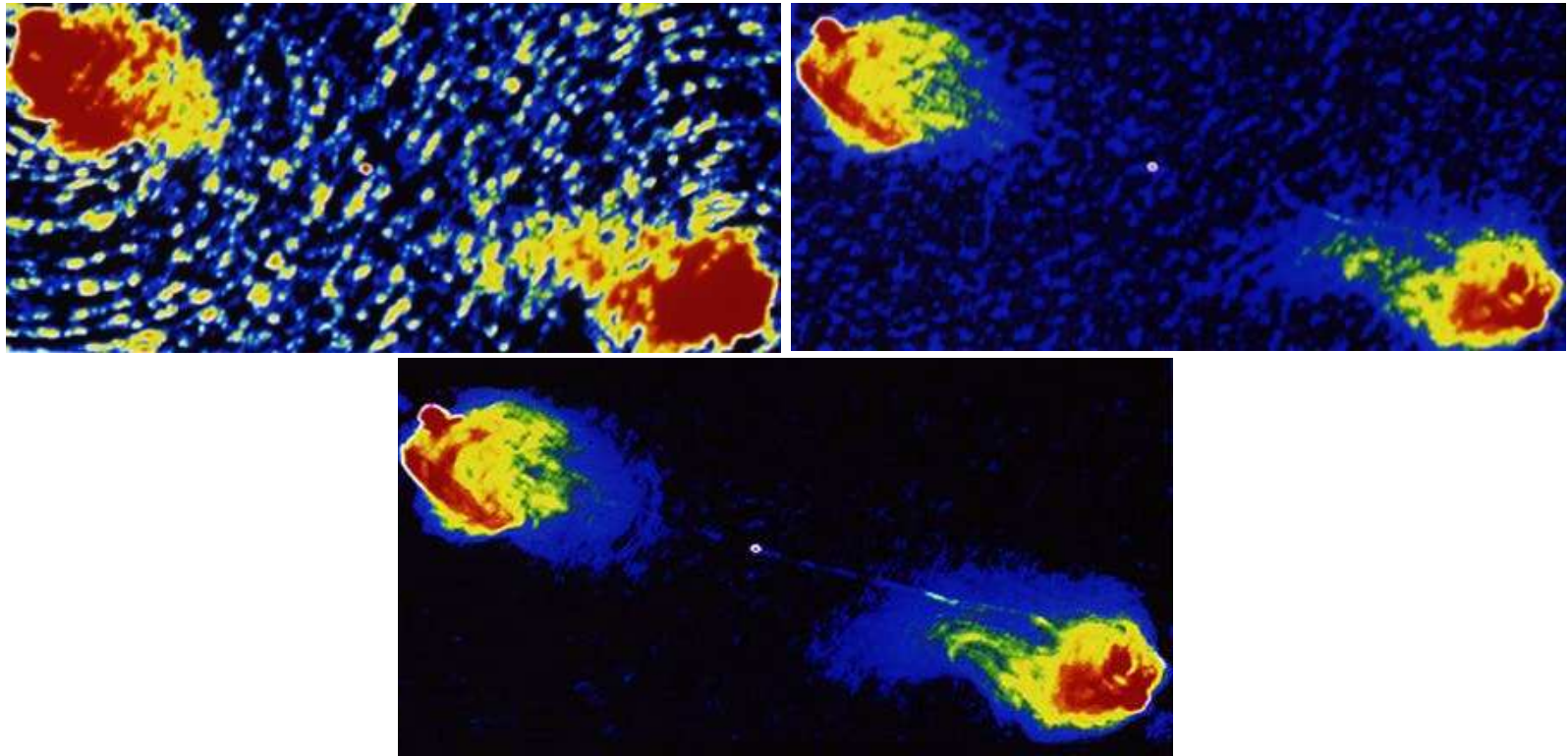
Example: VLA Zenith Snapshot



Example: VLA UV coverage

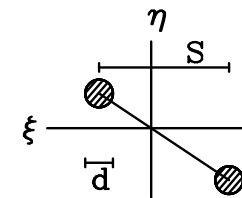
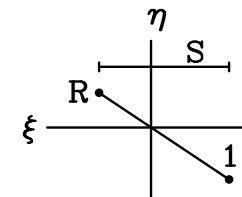
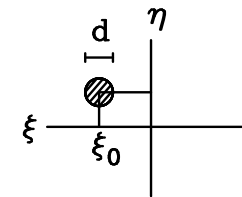
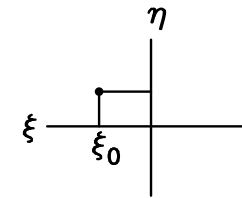
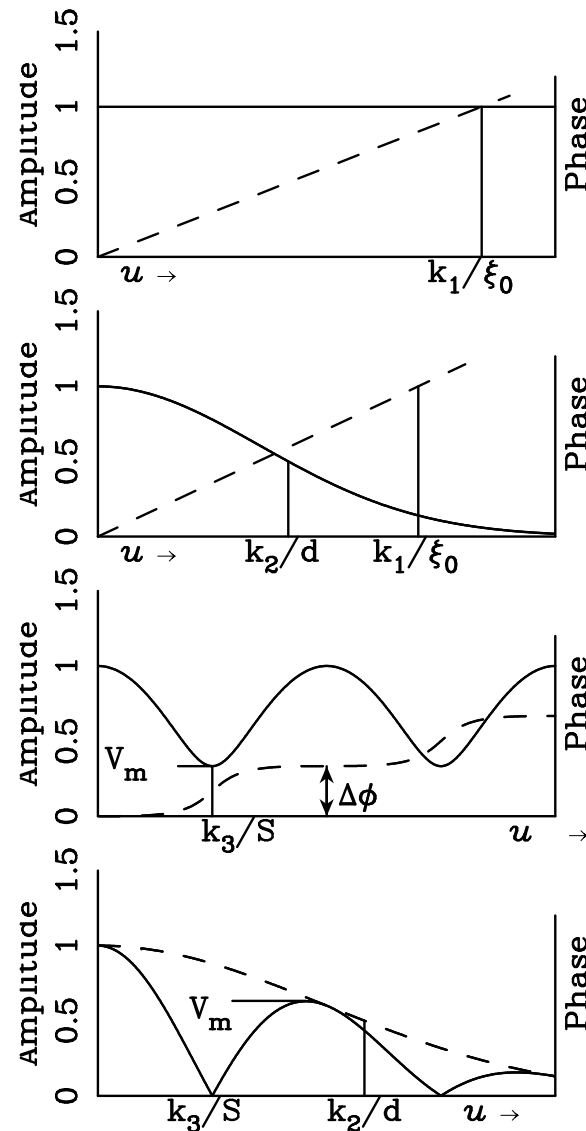


Example: Cygnus A by VLA



Three images of the radio galaxy Cygnus A, taken with the VLA at 6 cm (Perley, Dreher, courtesy of NRAO/AUI). **Top left:** Gridded and Fourier Transformed, with no additional processing. **Much of the structure is not real**, but rather due to side lobes of the synthesized beam. **Top right:** the image was deconvolved using the MEM method. Note the vast increase in dynamic range. The image in the bottom panel has been self-calibrated. This gives an additional factor of 3 in dynamic range.

Radio Interferometer



Diffraction and interference patterns. Solid lines are Amplitudes. Dashed lines are phases.

Fourier Transform in 1 Dimension

Fourier Transform:

$$F(s) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi s x} dx$$

Inverse Fourier Transform:

$$f(x) = \int_{-\infty}^{\infty} F(s) e^{i2\pi x s} ds$$

Comparing to our notation, we have $x \rightarrow \xi$ and $s \rightarrow u$.

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a } F(\frac{s}{a})$
Addition	$f(x) + g(x)$	$F(s) + G(s)$
Shift	$f(x - a)$	$e^{-i2\pi a s}$
Convolution	$f(x) \otimes g(x)$	$F(s)G(s)$
Modulation	$f(x) \cos x$	$\frac{1}{2} F(s - \frac{w}{2\pi}) + \frac{1}{2} F(s + \frac{w}{2\pi})$

For more, please look up Appendix B in the book "Tools of Radio Astronomy".

Fourier Transform in 1 Dimension

For a point source at $\xi = 0$, we have $f(\xi) = \delta(\xi)$ and the Fourier Transform $F(u) = 1$. For a point source at $\xi = \xi_0$, we can use the [shift theorem](#) and

$$\begin{aligned} F(u) &= e^{-i2\pi u\xi_0} \equiv \mathcal{V}e^{i\phi_v} \\ &= \cos(2\pi u\xi_0) - i\sin(2\pi u\xi_0) \equiv V_R + iV_I \end{aligned}$$

Thus, the amplitude and the phase are

$$\begin{aligned} |\mathcal{V}| &= [\cos^2(2\pi u\xi_0) + \sin^2(2\pi u\xi_0)]^{1/2} = 1 \\ \phi_v &= \tan^{-1} \left[\frac{-\sin(2\pi u\xi_0)}{\cos(2\pi u\xi_0)} \right] = -2\pi u\xi_0 \end{aligned}$$

[The fringe amplitude does not depend on the position offset \$\xi_0\$](#) . So when $u = \frac{1}{\xi_0}$ or ($u = \frac{206265}{\xi_0(\text{arcsec})}$ if ξ_0 is in unit of arcsec), $\phi_v = -2\pi$ or the phase is shifted by one fringe. Therefore, a displacement of $\xi_0 = 1''$ shifts phase by one fringe for a 206265 wavelength baseline.

Fourier Transform in 1 Dimension

For a gaussian source at $\xi = 0$, we have a normalized function

$$f(\xi) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\xi^2}{2\sigma^2}\right)$$

where σ is rms size and the FWHM $d = \sigma\sqrt{8\ln 2}$.

The Fourier Transform of the Gaussian $f(\xi) = \exp(-\pi\xi^2)$ is the Gaussian $F(u) = \exp(-\pi u^2)$.

We can solve it with [similarity theorem](#). Let $a = 1/(\sqrt{2\pi}\sigma)$, we have $f(a\xi)$ and

$$\begin{aligned} F(u) &= \frac{1}{\sqrt{2\pi}d} \left\{ \frac{1}{a} \exp\left[-\pi\left(\frac{u}{a}\right)^2\right] \right\} \\ &= \exp[-2\pi^2 u^2 \sigma^2] \equiv V_R + iV_I \end{aligned}$$

It is clear that $V_I = 0$ because the Gaussian is a symmetric function about $u = 0$. Thus $\phi_v = 0$ and

$$|\mathcal{V}| = V_R = \exp[-2\pi^2 u^2 \sigma^2]$$

Thus $|\mathcal{V}| = 0.5$ when $u = \sqrt{\frac{\ln 2}{2\pi^2 \sigma^2}} = \frac{2 \ln 2}{\pi d}$

Fourier Transform in 1 Dimension

If the center of the Gaussian is shifted by $\xi = \xi_0$, so that we have

$$f(\xi) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\xi - \xi_0)^2}{2\sigma^2}\right]$$

then with the [shift theorem](#), we have

$$F(u) = \exp(-i2\pi u\xi_0) \exp[-2\pi^2 u^2 \sigma^2] \equiv V_R + iV_I$$

Thus, we have

$$|\mathcal{V}| = \exp[-2\pi^2 u^2 \sigma^2]$$

and

$$\phi_v = -2\pi u\xi_0$$

Again, the fringe amplitude does not depend on the position offset ξ_0 .

Fourier Transform in 1 Dimension

For two point sources, one at $\xi = \xi_0$ and one at $\xi = \xi_0 - S$ with $S > \xi_0 > 0$. If the two point source have an intensity ratio $R \geq 1$ and the centroid is the phase center, then we have $S = (R + 1)\xi_0$ and

$$f(\xi) = \frac{R\delta(\xi - \xi_0) + \delta(\xi - \xi_0 + S)}{R + 1}$$

With the "Addition" and "shift" theorems, we have

$$F(u) = \frac{Re^{-i2\pi u\xi_0} + e^{-i2\pi u(\xi_0 - S)}}{R + 1} \equiv V_R + iV_I$$

So, it a **vector sum of the two visibilities**. Thus, the amplitude and the phase are

$$\begin{aligned} |\mathcal{V}| &= \sqrt{F(u)F^*(u)} = \frac{1}{R + 1} \sqrt{R^2 + 1 + 2R \cos(2\pi uS)} \\ \phi_v &= -\tan^{-1} \left\{ \frac{R \sin(2\pi u\xi_0) + \sin[2\pi u(\xi_0 - S)]}{R \cos(2\pi u\xi_0) + \cos[2\pi u(\xi_0 - S)]} \right\} \end{aligned}$$

So the minimal $|\mathcal{V}|_{\min} = \frac{R-1}{R+1}$ when $u = \frac{1}{2S}$. The amplitude is shifted by one fringe when $u = \frac{1}{S}$ and $\phi_v = -\frac{2\pi}{R+1}$.

Fourier Transform in 1 Dimension

If the two point sources become Gaussians and $R = 1$, then we have $S = 2\xi_0$ and

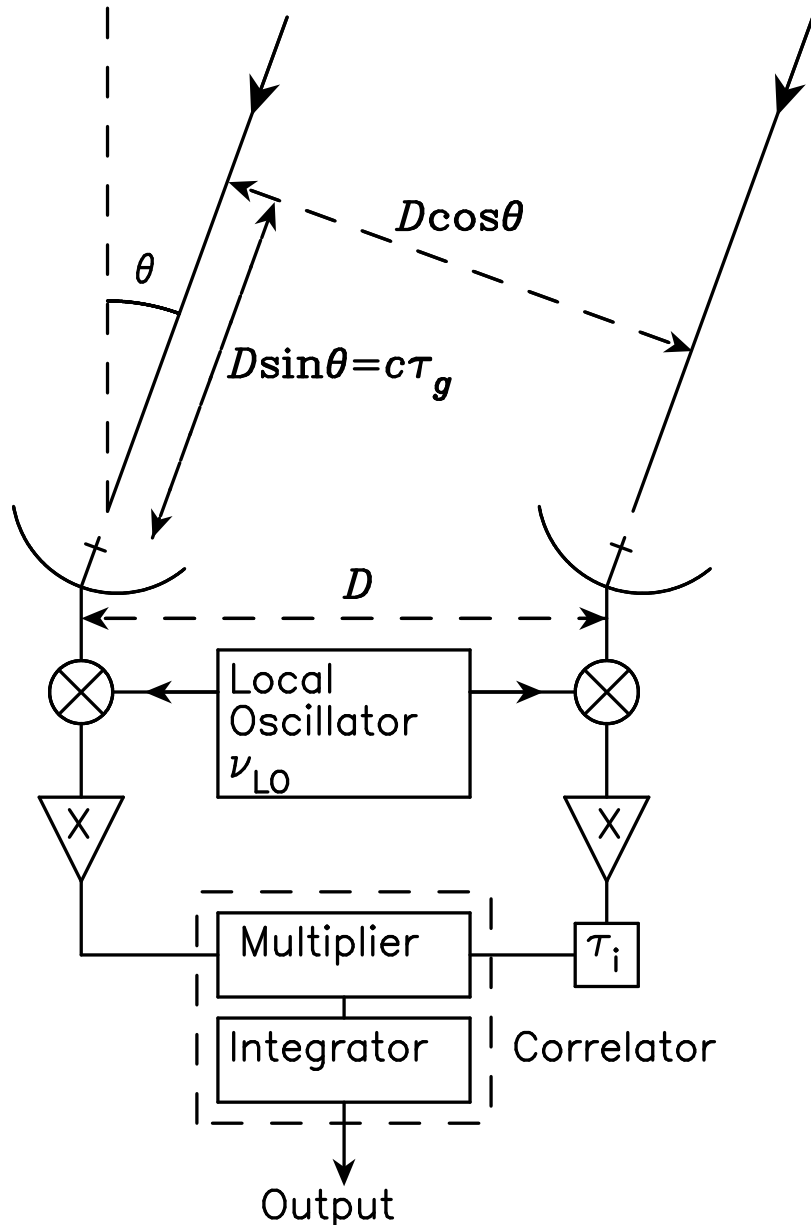
$$F(u) = \exp[-2\pi^2 u^2 \sigma^2] \frac{R e^{-i2\pi u \xi_0} + e^{-i2\pi u (\xi_0 - S)}}{R + 1} \equiv V_R + iV_I$$

Thus, the amplitude and the phase are

$$\begin{aligned} |\mathcal{V}| &= \sqrt{F(u)F^*(u)} = \exp[-2\pi^2 u^2 \sigma^2] \frac{\sqrt{2 + 1 + 2 \cos(2\pi u S)}}{2} \\ \phi_v &= -\tan^{-1} \left\{ \frac{\sin(2\pi u \xi_0) + \sin[2\pi u (\xi_0 - S)]}{\cos(2\pi u \xi_0) + \cos[2\pi u (\xi_0 - S)]} \right\} \end{aligned}$$

The amplitude is shifted by one fringe when $u = \frac{1}{S}$ and $\phi_v = -\frac{2\pi}{2}$. $\implies$ Two-slit interference

Radio Interferometer



In practice, all receiving systems in radio astronomy the frequencies of the signals received at the antennas are changed by **mixing with a local oscillator signal**. This feature, referred to as frequency conversion (or heterodyne frequency conversion), enables the major part of the signal processing to be performed at intermediate frequencies that are most appropriate for amplification, transmission, filtering, delaying, recording, and similar processes.

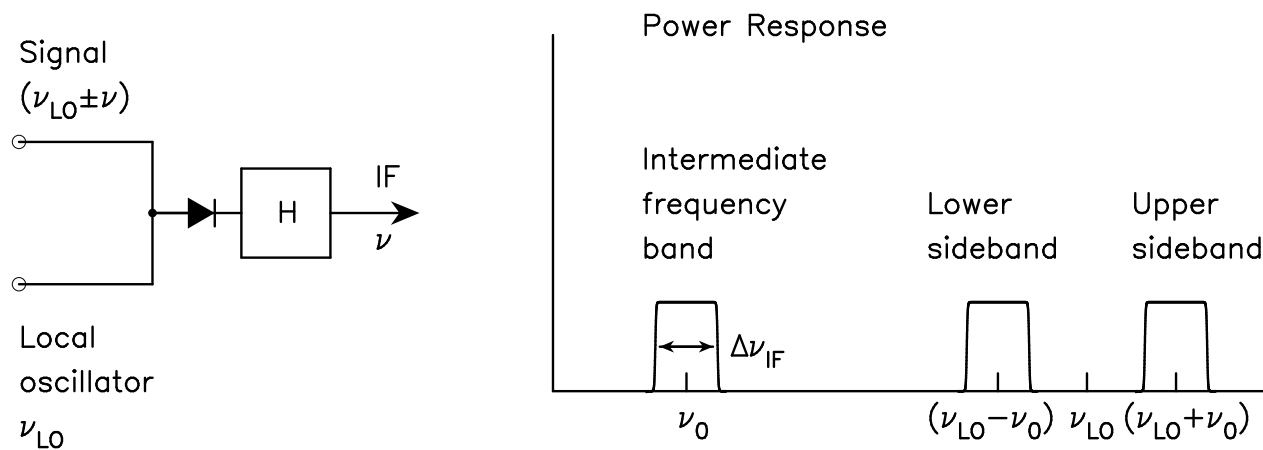
⊗ : Mixers

LO: Local Oscillator

Radio Interferometer

LO has a voltage of $b_1 \cos(2\pi\nu_{LO}t + \theta_{LO})$. Signal is $b_2 \cos(2\pi\nu_s t + \phi_s)$. So output of the mixer is:

$$\begin{aligned}
 & b_1 \cos(2\pi\nu_{LO}t + \theta_{LO}) \\
 & \times b_2 \cos(2\pi\nu_s t + \phi_s) = \frac{1}{2} b_1 b_2 \cos[2\pi(\nu_s + \nu_{LO})t + \phi_s + \theta_{LO}] \\
 & + \frac{1}{2} b_1 b_2 \cos[2\pi(\nu_s - \nu_{LO})t + \phi_s - \theta_{LO}]
 \end{aligned}$$



Left: Simplified diagram of a mixer and a filter H that defines the **intermediate frequency (IF) band centered on ν_0** . The nonlinear element shown is a diode. **Right:** Signals from within the bands centered on $\nu_{LO} - \nu_0$ (lower sideband) and $\nu_{LO} + \nu_0$ (upper sideband) are converted to the IF band and admitted by the filter. **For the SMA, IF (ν_0) is 5 GHz with a bandwidth $\Delta\nu_{IF} = 2$ GHz.** If one side band is needed, the other band can be removed by a suitable filter.

Noise

In practical systems, there will always be several sources of system noise in the receiver output:

1. the receiver
2. atmosphere
3. the spillover noise of the antenna which consists of ground radiation that enters into the system through far side lobes
4. noise produced by unavoidable attenuation in wave guides and coaxial cables connecting the feed to the receiver input

These noise contributions are **additive** and therefore

$$T_{\text{sys}} = \sum T_i$$

where T_i is the noise temperature of the individual contributors. If the total receiver bandwidth is $\Delta\nu$, then two samples taken at time intervals less than $\Delta t = 1/\Delta\nu$ apart are not independent. Then a total time τ contains $N = \tau/\Delta t = \tau \cdot \Delta\nu$ independent samples. Gaussian statistics show that the total error for N samples is $1/\sqrt{N}$ of that of single sample. Calling the error of this single sample T_{sys} , we obtain

$$\Delta T_A = \frac{T_{\text{sys}}}{\sqrt{t\Delta\nu}}$$

Interferometer Noise

The RMS fluctuations (noise) in antenna temperature are

$$\Delta T_A = \frac{MT_{\text{sys}}}{\sqrt{t\Delta\nu}}$$

where M is a factor of order unity used to account for extra noise from analog to digital conversions, digital clipping etc. For a single antenna, the flux density S_ν is related to the antenna temperature through the power (assuming one polarization) that receives:

$$W_\nu d\nu = \frac{1}{2} S_\nu A_e d\nu = kT_A d\nu \quad \Rightarrow \quad S_\nu = \frac{2kT_A}{A_e} = \Gamma^{-1} T_A \quad \text{with} \quad \Gamma = \frac{A_e}{2k}$$

where Γ is the system sensitivity. For a two-element system (i.e. interferometer), the fluctuation (noise) in antenna temperature and flux density, including the effect of atmosphere opacity τ , is:

$$\Delta T_A = \frac{MT_{\text{sys}}e^\tau}{\sqrt{2t\Delta\nu}} \quad \Rightarrow \quad \Delta S_\nu = \Gamma^{-1} \Delta T_A = \frac{2MkT_{\text{sys}}e^\tau}{A_e\sqrt{2t\Delta\nu}}$$

where A_e is the effective collecting area of a single telescope of diameter D . The extra factor of $\sqrt{2}$ arises from the use of 2 antennas each of collecting area A_e , and the fact that the correlation of two noisy signals (samples) leads to an increase in the noise by a factor of $\sqrt{2}$.

Interferometer Noise

To simplify the derivation, let's denote the system noise corrected for atmospheric absorption by

$$T'_{\text{sys}} = T_{\text{sys}} e^{\tau}$$

For an array of n identical telescopes, there are $N = n(n - 1)/2$ simultaneous pair-wise correlations. The the RMS variation in flux density is

$$\Delta S_{\nu} = \frac{2MkT'_{\text{sys}}}{A_e \sqrt{2Nt\Delta\nu}}$$

This relation can be recast in the form of brightness temperature T_b fluctuations using the Rayleigh-Jeans relation:

$$S_{\nu} = I_{\nu} \Omega_b \approx \frac{2kT_b}{\lambda^2} \Omega_b \quad \Rightarrow \quad T_b = \frac{\lambda^2}{2k\Omega_b} S_{\nu}$$

Then [the noise in brightness temperature](#), due to random noise, in aperture synthesis image is

$$\Delta T_b = \frac{M\lambda^2 T'_{\text{sys}}}{A_e \Omega_b \sqrt{2Nt\Delta\nu}}$$

Interferometer Noise

From the previous slide,

$$\Delta S_\nu = \frac{2MkT'_{\text{sys}}}{A_e \sqrt{2Nt\Delta\nu}}$$

$$\Delta T_b = \frac{M\lambda^2 T'_{\text{sys}}}{A_e \Omega_b \sqrt{2Nt\Delta\nu}}$$

Note $\Omega_b = 1.133\theta^2 \propto (\lambda/D)^2$ for a Gaussian beam, where D is the equivalent length of the baseline. A few comments:

1. ΔT_b decreases with decreasing wavelength.
2. ΔT_b decreases with increasing Ω .
3. ΔT_b decreases with increasing bandwidth $\Delta\nu$ for continuum measurements.
4. Multi-element interferometers are capable of producing images faster than single dishes with the same ΔT_b , due to n receivers in an interferometer system. If a single dish of equivalent collecting area and angular resolution also had n receivers, it would produce an image of the same area in the same amount of time.

Noise

Inserting numerical values, we have

$$\begin{aligned}\Delta S_\nu &= 1.0 \frac{MT_{sys'}}{A_e \sqrt{Nt\Delta\nu}} \text{ Jy} \\ \Delta T_b &= 14.0 \frac{M\lambda^2 T'_{sys}}{A_e \theta^2 \sqrt{Nt\Delta\nu}} \text{ K}\end{aligned}$$

where λ is in mm, t in hr, θ in arcsec, and $\Delta\nu$ in kHz and A_e in m^2 , θ in arcsec. Let's compare the performance of the VLA at 1.3cm with the BIMA millimeter array at 2.7mm. Let $M=1$ for both arrays.

Line measurement:

For **VLA**, $n = 27$, $A_e = 300 \text{ m}^2$, 4hr integration on source with $T_{sys} = 100 \text{ K}$ and $\Delta\nu = 7.8 \text{ kHz}$, in a 3" beam, $\Delta S_\nu = 3.2 \text{ mJy}$ or $\Delta T_b = 0.8 \text{ K}$. For **BIMA**, $n = 10$, $A_e = 14 \text{ m}^2$, 4hr integration, with $T_{sys} = 100 \text{ K}$, $\Delta\nu = 25 \text{ kHz}$, $\Delta S_\nu = 0.11 \text{ Jy}$ or $\Delta T_b = 1.2 \text{ K}$.

Continuum measurement:

For VLA, $\Delta\nu = 50 \text{ MHz}$, RMS noise is $40 \mu\text{m Jy}$. For BIMA, $\Delta\nu = 500 \text{ MHz}$, RMS noise is 0.8 mJy .